



Scalable Transactional Fraud Detection Pipeline & Analysis

An in-depth analysis of **4.97 million simulated financial transactions** to detect anomalous patterns, quantify financial impact, and equip stakeholders with actionable intelligence for optimizing fraud detection thresholds.

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PYTHON · SQL SERVER · POWER BI

Executive Summary

Project Objective

Analyze a massive, highly imbalanced dataset of **4.97 million simulated financial transactions** to detect anomalous patterns and calculate the financial impact of fraudulent behavior.

The primary goal is to equip stakeholders with actionable intelligence to optimize fraud detection thresholds — balancing the mitigation of financial loss against the operational costs of false positives.

Tools Used



Python

Pandas, NumPy for data wrangling



MS SQL Server

T-SQL for ingestion & analysis



Power BI

Interactive dashboard visualization

Methodology & Data Engineering

Real-world ledger data is inherently messy. To mimic a production environment, a synthetic dataset was engineered with deliberate inconsistencies, nulls, and duplicates — then cleaned and loaded into a relational database for analysis.

Data Wrangling (Python)

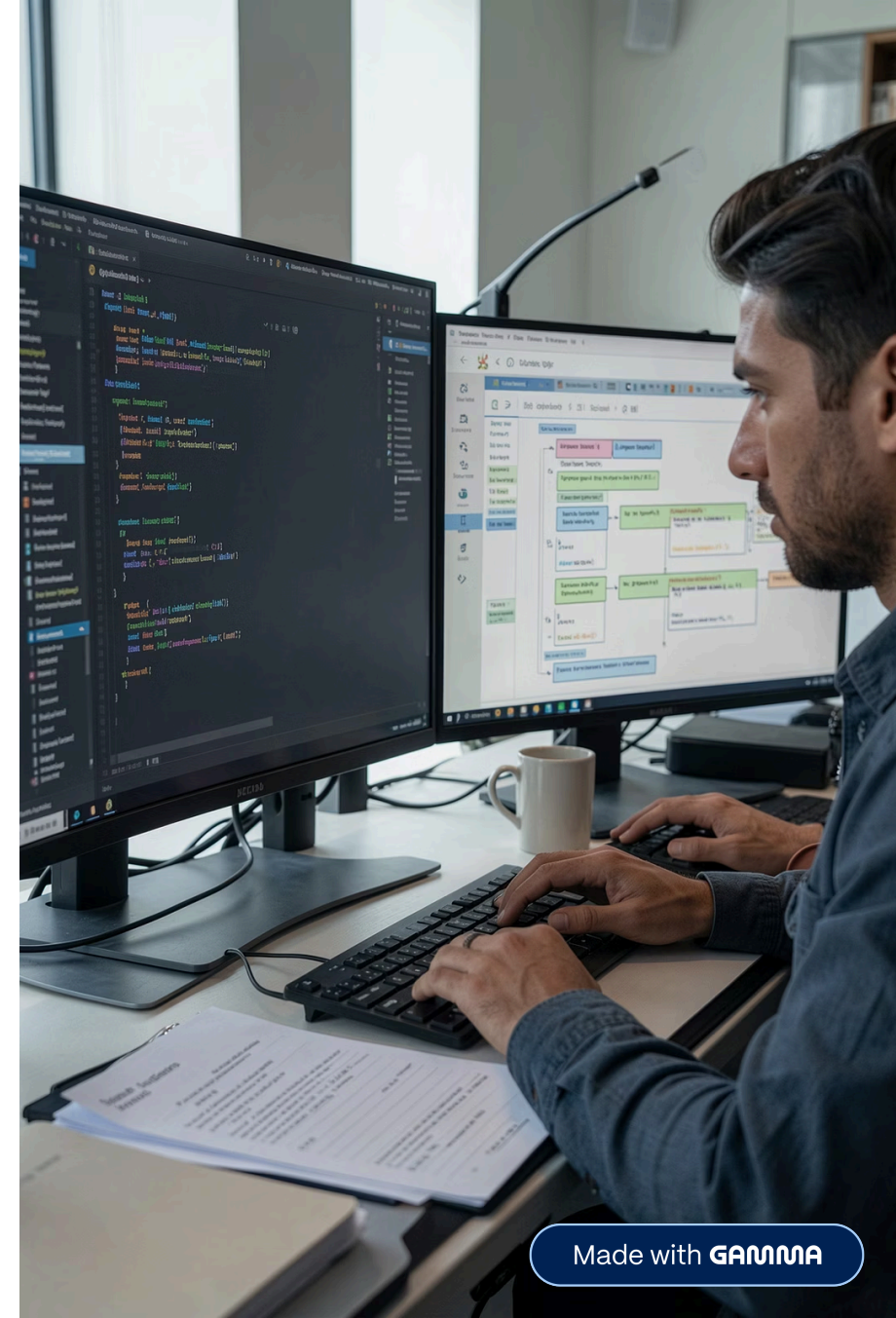
Executed deduplication on primary keys, imputed missing categorical values (merchant_category, payment_method), and standardized temporal and spatial string formats.

Data Ingestion (SQL Server)

Loaded the sanitized 4.97M row dataset into a relational database, enforcing NOT NULL constraints to ensure pipeline integrity.

Analysis (T-SQL)

Developed performant aggregation queries to extract KPIs related to transaction velocity, category vulnerabilities, and account compromises.



KPI 1: Macro-Level Health

Establishes the baseline cost of fraud. Even a fractional percentage of fraud at massive transaction scale translates to over **\$2 million in preventable business losses**.

4.97M

Total Transactions

Transactions analyzed in the dataset

50,115

Fraud Cases

Total fraudulent transactions identified

1.01%

Overall Fraud Rate

Percentage of transactions flagged as fraud

\$2.05M

Total Financial Loss

Preventable losses from fraudulent activity

📌 Total processed volume: **\$205,187,565.33** — highlighting the scale at which even a 1.01% fraud rate becomes a significant financial exposure.

KPI 2: Transaction Value Behavior

Key Findings

Average Legitimate Transaction

\$41.25 — typical everyday spend

Average Fraudulent Transaction

\$40.81 — fraudsters target slightly lower ticket sizes to avoid high-value alerts

Maximum Single Fraud Event

\$4,229.95 — outlier events demand hard-stop thresholds

Strategic Insight

Fraudsters in this dataset tend to target slightly lower, "everyday" ticket sizes compared to legitimate users to avoid triggering high-value alerts. However, the presence of outlier fraud events (up to \$4,229.95) indicates the need for **dual-layered thresholds**:

- **Velocity checks** for small, frequent amounts
- **Hard stops** for massive single-transaction amounts

KPI 3: Vulnerability by Payment Method

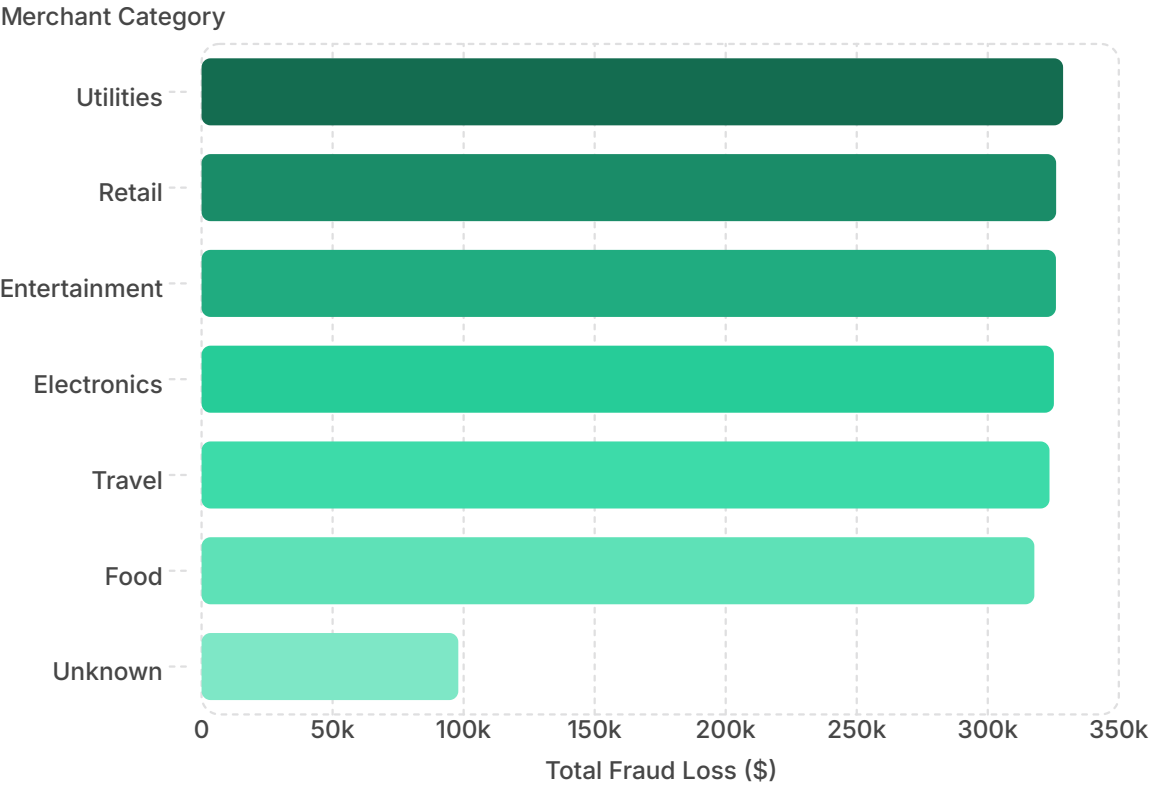
Credit Cards and NetBanking are the most exploited infrastructures, accounting for the **highest financial losses (over \$1M combined)** despite transaction volumes being nearly identical across all channels.

Payment Method	Total Transactions	Fraud Rate	Financial Loss
Credit Card	1,218,170	1.01%	\$515,742.82
NetBanking	1,219,062	1.01%	\$511,689.02
UPI	1,219,645	1.00%	\$491,422.88
Debit Card	1,218,648	1.00%	\$487,234.81
Unknown	99,475	1.02%	\$39,254.58

 **Recommendation:** Secondary authentication measures should be aggressively routed to Credit Card and NetBanking channels given their disproportionate financial exposure.

KPI 4: High-Risk Merchant Categories

Fraudsters are primarily cashing out through everyday categories like **Utilities** and **Retail**. Stricter transaction velocity rules should be deployed specifically for these merchant codes, as fraudsters appear to be testing stolen credentials on basic bills and retail goods.



Top 3 Exploited Categories

Utilities

\$328,738.33 lost

Retail

\$326,084.02 lost

Entertainment

\$326,007.37 lost

KPI 5: Most Compromised Accounts

Isolated the **Top 10 customer accounts** exhibiting the highest volume of fraudulent activity. These accounts represent immediate "fire drills" for the fraud operations team.

Customer ID	Total Txns	Fraud Txns	Total Fraud Amount
CUST_59196	42	2	\$4,233.70
CUST_39732	54	2	\$3,595.85
CUST_51675	45	1	\$1,760.28
CUST_18702	43	3	\$1,569.85
CUST_18846	46	1	\$1,562.06
CUST_32105	52	2	\$1,490.88
CUST_75449	60	1	\$1,441.06
CUST_85857	39	2	\$1,419.30
CUST_27015	44	2	\$1,362.16
CUST_20193	47	4	\$1,317.62

⊗ **Immediate Action Required:** These accounts require immediate freezing, automated credential resets, and customer outreach to prevent further financial bleeding.



Business Recommendations

1

Dynamic Thresholding

Implement stricter transaction limits (velocity checks) specifically on **Credit Card and NetBanking channels** (identified in KPI 3), rather than applying a blanket rule that inconveniences all users.

2

Targeted Merchant Monitoring

Deploy an automated alert system for rapid, back-to-back transactions occurring within the **Utilities and Retail sectors** (KPI 4), where fraudsters appear to be testing stolen credentials.

3

Automated Account Freezes

Utilize the logic from KPI 5 to automatically suspend accounts that trigger multiple anomaly flags within a **24-hour window** (or exceed single-transaction limits of \$1,000+), routing them to human agents for manual review.

Power BI Dashboard: Fraud Analytics & Risk Optimization

An interactive dashboard was built in Power BI to represent the insights visually, providing a comprehensive overview of fraud metrics for stakeholder decision-making.

Key Dashboard KPIs

1.01%

Overall Fraud Rate

\$2.05M

Total Fraud Loss

\$205.19M

Total Volume Processed

Visualizations

- **High-Risk Merchants:** Horizontal bar chart showing total fraud loss by merchant category (Utilities ~\$0.32M, Retail ~\$0.31M, Entertainment ~\$0.30M)
- **Payment Method Vulnerability:** Donut chart illustrating fraud loss distribution — Credit Card 25.22%, NetBanking 25.02%, UPI 24.03%, Debit Card 23.82%, Unknown 1.92%